

Week:8

Experiment:5

“To determine the amount of carbon dioxide exhaled while resting and during activity.”

Total Marks (30 Marks):

1) Lab Performance (10 Marks):

2) Observation (10 Marks):

Condition	Initial Readings (mL)	Final Readings (mL)	Volume of NaOH used (mL)	Time for Decolorization (s)	Moles of CO ₂ Exhaled
Rest	0.2	0.9	0.7	25.58s	1.4×10^{-3}
Exercise	1.0	2.7	1.7	7.76s	3.4×10^{-3}

3) Calculations (5 Marks):

- 1ml of 0.1 M NaOH combines with $= 5 \times 10^{-5}$ moles of Carbon dioxide
- X ml of 0.1 M NaOH combines with $= 5 \times 10^{-5} \times X$ moles of Carbon dioxide
- **Rest:** $X = 0.7\text{mL}$

$$\text{moles} = 5 \times 10^{-5} \times 0.7 = 0.000035 = 3.50 \times 10^{-5} \text{mol}$$

- **Exercise:** $X = 1.7\text{mL}$:

$$\text{moles} = 5 \times 10^{-5} \times 1.7 = 0.000085 = 8.50 \times 10^{-5} \text{mol}$$

Mass of CO₂:

$$\text{mass} = \text{moles} \times 44.01 \text{ g/mol}$$

- **Rest:** $3.50 \times 10^{-5} \times 44.01 = 0.001540\text{g} = \mathbf{1.540 \text{ mg}}$
- **Exercise:** $8.50 \times 10^{-5} \times 44.01 = 0.003741\text{g} = \mathbf{3.741 \text{ mg}}$

Volume of CO₂ at STP

volume (L) = moles $\times$ 22.414 L/mol

- **Rest:** $3.50 \times 10^{-5} \times 22.414 = 7.8449 \times 10^{-4} \text{L} = \mathbf{0.7845 \text{ mL}}$
- **Exercise:** $8.50 \times 10^{-5} \times 22.414 = 1.9052 \times 10^{-3} \text{L} = \mathbf{1.9052 \text{ mL}}$

Rate (moles of CO₂ per second)

rate (mol/s) = moles $\div$ time

- **Rest:** $3.50 \times 10^{-5} \div 25.58 = 1.368 \times 10^{-6} \text{mol/s}$
- **Exercise:** $8.50 \times 10^{-5} \div 7.76 = 1.096 \times 10^{-5} \text{mol/s}$

Formula:

$$\mathbf{X \text{ ml contains moles of CO}_2 = 5 \times 10^{-5} \times X}$$

4) Result (5 Marks):

The experiment shows that the amount of carbon dioxide exhaled increases after physical activity. During rest, 0.7 mL of NaOH was neutralized, corresponding to 3.5×10^{-5} mol of CO₂, whereas during exercise, 1.7 mL of NaOH was used, giving 8.5×10^{-5} mol of CO₂. Exercise enhances the rate of respiration, resulting in greater carbon dioxide production and faster decolorization time.

6) Problems:

Small errors in reading burette or pipette volumes could affect CO₂ calculation accuracy.

Reaction time in starting or stopping the stopwatch can lead to slight errors.